RAJALAKSHMI ENGINEERING COLLEGE

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M2

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|---|---|
| **Started on** | Thursday, 13 August 2026, 10:54 AM |
| **State** | Finished |
| **Completed on** | Thursday, 13 August 2026, 10:57 AM |
| **Time taken** | 3 mins 10 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

**Note:** No need of counter increment for declarations and scanf() and counter variable printf() statement.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**Answer:**

```
#include<stdio.h>
int count=0;
 void Factor(int num)

 {
     count++;
    for (int i = 1; i <= num;++i)
    {count++;
     if (num % i== 0)
        {count++;

        }
        count++;
     }

  }
  int main(){
      int num;
      scanf("%d",&num);
      Factor(num);
      printf("%d",count);
  }
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 12 | 31 | 31 | ✔ |
| ✔ | 25 | 54 | 54 | ✔ |
| ✔ | 4 | 12 | 12 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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